

Gaussian Splatting Feature Fields for (Privacy-Preserving) Visual Localization

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Abstract

Visual localization is the task of estimating a camera pose in a known environment. In this paper, we utilize 3D Gaussian Splatting (3DGS)-based representations for accurate and privacy-preserving visual localization. We propose Gaussian Splatting Feature Fields (GSFFs), a scene representation for visual localization that combines an explicit geometry model (3DGS) with an implicit feature field. We leverage the dense geometric information and differentiable rasterization algorithm from 3DGS to learn robust feature representations grounded in 3D. In particular, we align a 3D scale-aware feature field and a 2D feature encoder in a common embedding space through a contrastive framework. Using a 3D structure-informed clustering procedure, we further regularize the representation learning and seamlessly convert the features to segmentations, which can be used for privacy-preserving visual localization. Pose refinement, which involves aligning either feature maps or segmentations from a query image with those rendered from the GSFFs scene representation, is used to achieve localization. The resulting privacy- and non-privacy-preserving localization pipelines, evaluated on multiple real-world datasets, show state-of-the-art performances.

1. Introduction

Visual localization (VL), a core part of self-driving cars [26] and autonomous robots [36], is the task of estimating the 6DoF camera pose from which an image was captured.

VL methods can be distinguished by how they represent scenes and how they estimate the pose of a query image w.r.t. the scene representation. Popular choices for the former include 3D Structure-from-Motion (SfM) point clouds [23, 28, 29, 56, 57], databases of images with known intrinsic and extrinsic parameters [2, 59, 86, 89], the weights of neural networks [4, 6, 7, 32, 45], or dense renderable rep-

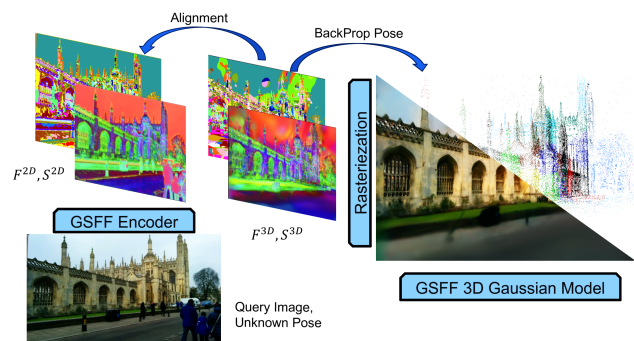


Figure 1. We extract 2D feature $\mathbf{F}^{2D}$ or segmentation $\mathbf{S}^{2D}$ maps from a query image with unknown pose. We estimate the pose by aligning the maps with feature $\mathbf{F}^{3D}$ or segmentation $\mathbf{S}^{3D}$ maps rendered from our Gaussian Splatting Feature Fields (GSFFs), which are learned in a self-supervised way. Using segmentations instead of features makes our localization pipeline privacy-preserving.

resentations such as meshes [51, 70], neural radiance fields [13, 46, 53, 91], or 3D Gaussian Splatting [3, 39, 64, 84]. Common VL approaches are based on establishing 2D-3D correspondences via feature matching [46, 51, 56, 67, 91] or scene coordinate regression [4, 6], relative pose estimation from feature matches [2, 86, 88, 89] or regression [47, 89], absolute pose regressed via neural networks [11, 12, 45, 74], or feature-based pose refinement [52, 57, 70] of an initial pose estimate. While local feature matching-based approaches provide the most accurate camera pose estimates, scale to large scenes [29, 58], handle changing conditions [68] and fit onto mobile devices [36, 41], they suffer from potential privacy-related issues as image details can be recovered from the feature descriptors [9, 55].

Since VL systems are often deployed through cloud-based solutions, preserving the privacy of user-uploaded images and scenes is a critical aspect [65, 66]. An interesting approach to both query and scene privacy is SegLoc [52], which is a pose refinement-based method that represents the scene as a sparse SfM point cloud. Most

refinement-based methods project features associated with the scene geometry into the query image [46, 53, 57, 70, 72, 73]. An initial pose estimate, *e.g.*, obtained via image retrieval, is then optimized by aligning the projected features with features extracted from the query image. To increase privacy while decreasing memory requirements for storing high-dimensional features, SegLoc [52] propose to quantize the features into integer values, which is equivalent to assigning segmentation labels to pixels in the query image and the 3D scene points. These quantized representations leads to better privacy-preservation: only coarse image information without any details can be recovered from both 2D segmentation images and 3D point clouds with associated labels. Final poses are refined by maximizing label consistency between the projected points and pixel labels.

The segmentations used by SegLoc are learned purely in 2D, providing no guarantee that the predicted labels are consistent between viewpoints. In contrast, [53] jointly trains a dense scene representation (in the form of a NeRF) together with an implicit feature field, ensuring multi-view consistency and state-of-the-art feature-based pose refinement results. Yet, [53] is not privacy-preserving. In contrast, in this work, we investigate learning a feature field that can be used for privacy-preserving visual localization jointly with a dense scene representation. We chose 3D Gaussian Splatting (3DGS) [33] due to its fast rendering time, making it particularly suited for dense pose refinement. Compared to SegLoc, it allows us to ground representation learning in 3D. Additionally, the explicit and finite nature of 3DGS is better suited for privacy-preserving localization than NeRFs as, similarly to [52], we can simply quantize the features associated with the Gaussians.

Some prior [3] and concurrent works [39, 64, 84] also use 3DGS in the context of visual localization. However, their pipelines use matching-based solutions, whereas we take advantage of the ability to densely render from any viewpoint within the scene to perform feature-metric or segmentation-based pose refinement. We explicitly back-propagate through the rasterizer to the $se(3)$ Lie algebra, which yields more accurate pose estimates. Furthermore, instead of relying on pre-trained features as [70], we learn our features in a self-supervised manner defining a Gaussian Splatting Feature Field (GSFFs), which associates 3D Gaussians with volumetric features extracted with a kernel-based encoding based on the covariance of the 3D Gaussians. These features are rendered and aligned to features provided by the 2D encoder (which is jointly trained with the GSFFs) through contrastive losses. We apply further regularization by leveraging the geometry of the 3DGS model and by spatially clustering the GSFFs. These clusters enables converting features into segmentations that can be used to perform effective privacy-preserving pose refinement (*c.f.* Fig. 1).

To summarize, our contributions are threefold: 1) We

introduce Gaussian Feature Fields, a novel representation for VL, jointly learnt with the image feature encoder enabling pose refinement by aligning rendered and extracted features. 2) The explicit nature of the 3D representation in GSFFs allows to spatially cluster the Gaussian cloud and segmenting the feature field based on cluster centers yielding a privacy-preserving 3D scene representation. Similar to [52], the resulting method is based on aligning discrete segmentation labels, thus leading to similar privacy-preserving properties. 3) We demonstrate the accuracy of both GSFFs-based localization pipelines (based on features respectively segmentations) on multiple real-world datasets. Our approaches outperform prior and concurrent work for privacy- and non-privacy-preserving visual localization.

2. Related Works

Gaussian Splatting. 3D Gaussian Splatting [33] represents a scene as a set of Gaussian primitives and render images by rasterizing Gaussians using depth ordering and alpha-blending, thus allowing efficient training and high resolution real-time rendering. Thanks to these properties, 3D Gaussians have been applied to numerous tasks including SLAM [31, 43], dynamic scene modeling [40, 78], scene segmentation [30, 35], and surface reconstruction [25, 27, 83]. Extensions have been tackling anti-aliasing [82], sparse view setups [76, 81], removing reliance on estimated poses [79], and regularization [14, 15, 87]. In this paper, we adopt the Gaussian Opacity Fields model [83] as our 3D representation, which integrates the anti-aliasing module from [82] and offers highly accurate geometry through regularization, critical for visual localization.

Camera pose refinement-based visual localization. Pose refinement approaches minimize, with respect to the pose, the difference between the query image and a rendering obtained by projecting the scene representation into the image using the current pose estimate [1, 20, 21, 60, 61]. The pose is iteratively refined from an initial estimate. L2 distances between deep features are often used to measure the differences between the query image and the projected features from the scene [24, 38, 57, 72, 73, 77]. [70] integrates pre-trained features with a particle filter, thus removing the need to learn representations per scene. [52] replaces deep features with segmentation labels and aligns them with 2D segmentation maps. In contrast, our GSFFs Pose Refinement pipeline (GSFFs-PR) associates a deep feature or a segmentation label to each 3D Gaussian primitive and performs pose refinement by 1) rasterizing these quantities and 2) explicitly backpropagating the feature-metric or segmentation errors through the rasterizer with respect to the camera pose.

Rendering-based visual localization. Leveraging the capability of novel view synthesis of NeRFs, iNeRF [37, 80]

is the first to perform pose refinement through photometric alignment, where the optimization is performed by direct gradient descent back-propagation through the neural field. [37, 42, 70] parallelize this process with faster neural radiance fields or particle filters. NeRF-based representations have used to match [46, 91] or align rendered features from an implicit field [13, 53].

Multiple visual localization works have emerged leveraging the efficient 3DGS models instead of NeRFs. [3] matches rays emerging from ellipsoids with pixels with an attention mechanism and estimates the pose with a closed form solution. [39] performs matching between query and rendered image, lifting the matches in 3D with the rendered depth to refine the pose by solving a PnP problem. Similar to NeRF-based approaches, [64, 84, 85] distills pretrained features in a Gaussian-based 3D feature field. These methods estimate poses by matching these features followed by PnP+RANSAC. In contrast, we do not rely on pretrained features but learn our representations in a self-supervised manner. Furthermore, we solely estimate query poses through feature-metric refinement by explicitly back-propagating through the rasterizer, which yields higher accuracy even without additional PnP+RANSAC step. Furthermore, contrarily to concurrent works our 3D features take scale into account.

Privacy-preserving visual localization. Pittaluga et al. [55] have shown that detailed and recognizable images of a scene can be obtained from sparse 3D point clouds associated with local descriptors. While, [62, 65, 66] tried to address this by transforming 3D point clouds into 3D line clouds, it was shown in [8, 34] that these line clouds still preserve a significant amount of information about the scene geometry that can be used to recover 3D point clouds, hence enabling the inversion attack [55]. Instead, [34] lift pairs of points to lines and [44] relies on a few spatial anchors resulting in 3D lines with non uniformly sampled distributions. Another strategy is to swap coordinates between random pairs of points [50] or performing partial pose estimation against distributed partials maps [22], but as shown in [9], the point positions may be recovered by identifying points neighborhoods from co-occurring descriptors.

Another line of work studies the privacy on the feature representation level. [19, 54] embed descriptors in subspaces designed to withstand different privacy attacks. Closer to our work, [52, 75, 90] remove the need for high-dimensional features. [75, 90] match keypoints and 3D points without descriptors and [52] replace high-dimensional descriptors with segmentation labels. Similar to [52], we also rely on labels to ensure privacy, but our dense label rendering through 3DGS achieves better localization accuracy than [52], which relies on reprojected sparse labels.

3. Gaussian Splatting Feature Fields (GSFFs)

First in Sec. 3.1 we introduce our Gaussian Splatting Feature Field (GSFFs) which associates a scale aware feature to each 3D Gaussian. We describe how to jointly optimize it along with a 2D feature extractor by aligning rendered and 2D extracted features in a self-supervised manner. In Sec. 3.2 we propose to further improve the features’ discriminativeness and their 3D awareness, by clustering the 3D Gaussian cloud and encouraging that both features in a pixel aligned pair of 2D/3D features are close to the same prototype through an auxiliary contrastive loss. These prototypes not only help distilling spatial information in the feature space, but also allow for a smooth transition from feature maps to segmentation maps, enabling privacy-preserving localization (as described in Sec. 4). Finally, in Sec. 3.3 we describe the *GSFFs-PR Feature* visual localization pipeline that is based on pose refinement through feature alignment. Its extension, the privacy-preserving *GSFFs-PR Privacy* visual localization pipeline based on segmentations is described in Sec. 4. We show an overview of the architecture and training pipeline in Fig. 2.

3.1. Scene Representation

We adopt the Gaussian Opacity Fields model [83] as our Gaussian Field representation, where a scene is composed of a set of 3D Gaussian primitives parametrized by their center, scaling and rotation matrices, opacity and spherical harmonics coefficients. Given a ray r emanating from a camera pose P , [83] finds the intersection between the ray and 3D Gaussians and computes the contribution $C_i(r, P)$ of each Gaussian $\mathcal{G}_i$ traversed by the ray. The Gaussians are then ordered based on depth and the pixel’s color is obtained by alpha blending

$$c(r, P) = \sum_{i=1}^N c_i \alpha_i C_i(r, P) \prod_{j=1}^{i-1} (1 - \alpha_j C_j(r, P)) \quad , \quad (1)$$

where c_i is the view-dependent color, modeled with spherical harmonics associated with $\mathcal{G}_i$ and α_i are the blending weights. The ray-Gaussian intersection formulation allows for depth distortion and normal consistency regularization [83], which improves the geometry of the 3D scene.

3D feature fields. Similar to Eq. (1), features or segmentation labels can be rendered through the same alpha blending by simply replacing each Gaussian’s color by the feature or segmentation label assigned to the Gaussian. However, this requires that we assign to each Gaussian $\mathcal{G}_i$ a feature, which in general might be extremely costly, especially for high-dimensional features. To avoid this, instead of associating independent features to each Gaussian, we introduce a feature field parametrized by a triplane grid [10]. It is centered at the origin of the world coordinate

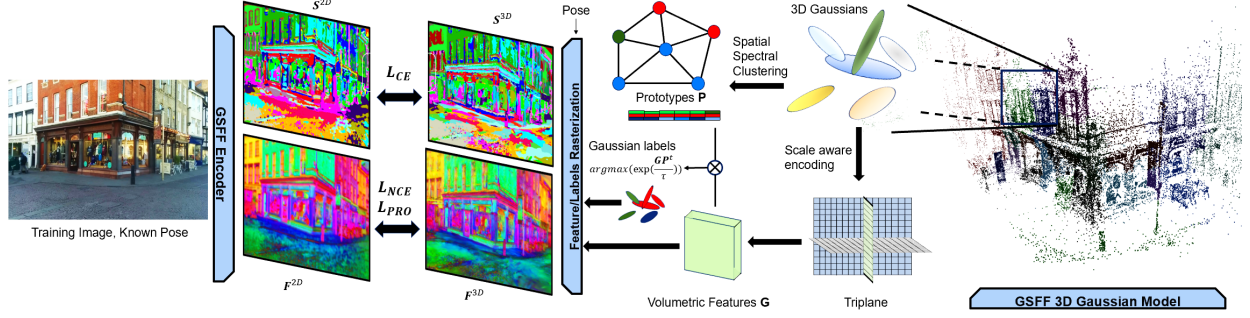


Figure 2. **Training pipeline.** We extract a feature map $\mathbf{F}^{2D}$ and segmentation map $\mathbf{S}^{2D}$ from a training image with known pose (left). For each 3D Gaussian $\mathcal{G}_i$ (right), a scale aware feature $\mathbf{g}_i$ is extracted from a triplane representation. Spectral clustering is applied on the Delaunay graph derived from the Gaussian cloud, yielding a set of prototypes P . A label is associated with each 3D Gaussian by assigning the volumetric features to the prototypes. The features and labels are then rendered to obtain the feature map $\mathbf{F}^{3D}$ and the segmentation map $\mathbf{S}^{3D}$, which are respectively aligned with their encoder counterparts $\mathbf{F}^{2D}$ and $\mathbf{S}^{2D}$ through L_{NCE} , L_{PRO} and L_{CE} .

space and is composed of three two-dimensional orthogonal planes $H_{xy}, H_{xz}, H_{yz} \in \mathbb{R}^{R \times R}$, where R is the resolution of the grid. For simplicity, we will also denote by $H_{xy}, H_{xz}, H_{yz} \in \mathbb{R}^{R \times R \times D}$ (and use them interchangeably) the three corresponding feature tensors, where D is the dimensionality of the feature space we aim to learn.

To compute the volumetric feature $\mathbf{g}_i$ associated with a 3D Gaussian $\mathcal{G}_i$ from the triplane, we proceed as follows. We project the 3D Gaussian $\mathcal{G}_i$ onto the three planes and compute for the three resulting 2D Gaussians $\mathcal{G}_i^{xy}, \mathcal{G}_i^{xz}, \mathcal{G}_i^{yz}$ three corresponding features $\mathbf{g}_i^{xy}, \mathbf{g}_i^{xz}, \mathbf{g}_i^{yz}$ by applying an RBF kernel parametrized by the 2D Gaussians (as detailed in the Supplementary). The three grid features are averaged yielding the volumetric feature $\mathbf{g}_i^{3D}$ associated with the Gaussian $\mathcal{G}_i$. This representation, called Gaussian Splatting Feature Field (GSFFs), enables the features to be scale-aware, as 3D Gaussians with large spatial span will aggregate feature information over a large area within the grid, and small Gaussians over smaller areas. Further advantages of the GSFFs representation are: 1) they allow sharing information between Gaussians based on overlapping projections onto the planes as the Gaussian features are optimized interdependently through rasterization, and 2) the field can be queried from any 3D position making it suitable for 3DGS splitting and merging mechanisms.

Self-supervised training. Our aim is to perform pose refinement based on aligning pixel-level 2D encoded features from the image with 3D rendered features. Hence it is important for these feature to be locally discriminative and robust to viewpoint changes. Formally, we want to learn the GSFFs feature field jointly with a 2D encoder such that when we render, at pose P , a feature map $\mathbf{F}^{3D}$ from the feature field, it is aligned with the corresponding 2D feature map $\mathbf{F}^{2D}$ of the image I associated to the pose P . The rendered feature map $\mathbf{F}^{3D}$ is obtained with alpha blending similarly to Eq. (1), where for each pixel u , we replace c_i

with $\mathbf{g}_i^{3D}$. To align $\mathbf{F}^{3D}$ and $\mathbf{F}^{2D}$, we train the model in a self-supervised manner with the contrastive loss [48]:

$$L_{NCE} = -\frac{1}{2HW} \sum_{u \in I} \log \left(\frac{\exp(\mathbf{F}_u^{3D} \cdot \mathbf{F}_u^{2D} / \tau)}{A} \right), \quad (2)$$

where τ is the temperature parameter, and A is a normalizing factor (detailed in the Supplementary).

3.2. Prototypical Feature Regularization

To better align the features and prepare the transition from features to segmentations, we propose to structure the feature space around a set of classes. To that end, we cluster the feature field, derive 3D prototypes, and apply an auxiliary contrastive loss to enforce that corresponding pixel-aligned 2D extracted and 3D rendered features are close to the same prototypes in the feature space.

Spatial prototypes. We want to find a set of feature prototypes that best encode the spatial prior reflecting the Gaussian cloud structure. One option would be to apply spectral clustering on a matrix containing pairwise distance between Gaussian centers. However given the high number of Gaussians, this would quickly become untractable. Therefore, we start with applying a Delaunay triangulation [17] of the Gaussian centers, which yields a graph that already captures local geometric information. Furthermore, the eigenvalues and eigenvectors can now be computed from the Laplacian of the sparse adjacency matrix derived from the graph. The resulting eigenvectors, one for each Gaussian $\mathcal{G}_i$, are clustered into K groups, which assigns each 3D Gaussian $\mathcal{G}_i$ to a cluster k . To build the representative feature (*cluster prototype*) $\mathbf{p}_k$ for each cluster, we simply average the volumetric features $\mathbf{g}_i$ of all Gaussians belonging to cluster k .

Prototypical loss. These prototypes implicitly define "classes" and we use them to maximize intra-class compactness and inter-class separability within the feature space

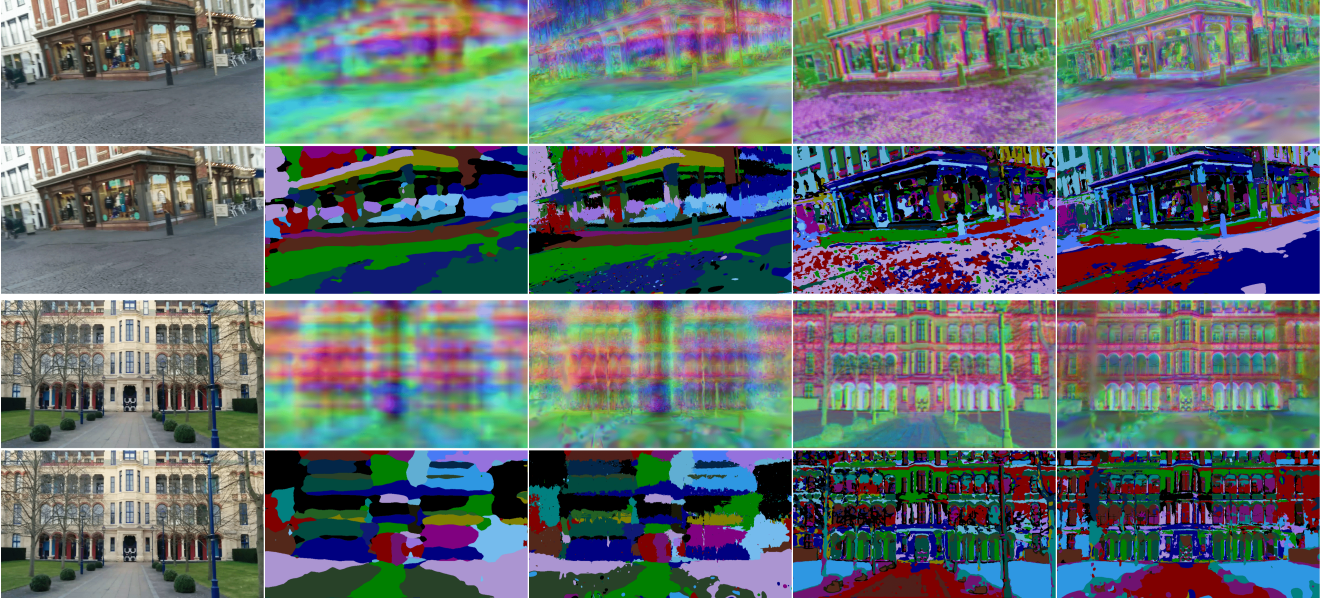


Figure 3. First and third lines: original image, encoder feature map $\mathbf{F}^{2D}$ (second and forth) and rendered feature map $\mathbf{F}^{3D}$ (third and fifth) at coarse level (second and third) and fine level (forth and fifth). Below (second and forth lines) correspondingly rendered image, encoder segmentation maps $\mathbf{S}^{2D}$ (second and forth) and rendered segmentation maps $\mathbf{S}^{3D}$ (third and fifth) at coarse and fine level respectively. Features are visualized through PCA downprojection.

and infuse spatial priors. Given a batch of N pairs of pixel-aligned rendered/encoder features $\{\mathbf{F}_u^{3D}, \mathbf{F}_u^{2D}\}$, we associate one prototype $\mathbf{p}_k$ per feature pair and want to enforce both features to be close to the prototype. We thus add the following prototypical contrastive loss:

$$L_{PRO} = -\frac{1}{N} \sum_{n=1}^N \log \left(\frac{\exp(\mathbf{F}_n^{3D} \mathbf{p}_n^t + \mathbf{F}_n^{2D} \mathbf{p}_n^t / \tau)}{B} \right), \quad (3)$$

where $\mathbf{F}_n^{2D}, \mathbf{F}_n^{3D}$ are the features corresponding to the pixel u_n , $\mathbf{p}_n$ is the prototype assigned to them, τ is a temperature parameter and B is a normalizing factor (detailed in the Supplementary along with derivations). To compute the associations over the batch of feature pairs, we use an optimal transport procedure based on the Sinkhorn-Knopp algorithm [16] (see the Supplementary for details).

Multi-view consistency. The rendering depends on the incidence of the ray traversing the 3D Gaussians. Hence rendered features may vary across viewing angles. In order to encourage multi-view consistency of both the encoder and the feature field, and to make our features generalizable to out-of-distribution views, we align features belonging to different views but associated to the same 3D points. To that end, given a pixel correspondence (u, v) between two images I and $\hat{I}$ in the scene – along with their encoder/rendered features $\{\mathbf{F}_u^{3D}, \mathbf{F}_u^{2D}\}, \{\hat{\mathbf{F}}_v^{3D}, \hat{\mathbf{F}}_v^{2D}\}$ – we randomly replace in L_{NCE} and L_{PRO} a subset of the pixel aligned feature pairs $\{\mathbf{F}_u^{3D}, \mathbf{F}_u^{2D}\}$ by $\{\hat{\mathbf{F}}_v^{3D}, \mathbf{F}_u^{2D}\}$ or $\{\mathbf{F}_u^{3D}, \hat{\mathbf{F}}_v^{2D}\}$. To extract correspondences, given a training image I with pose P and rendered depth D , we generate a

random pose $\hat{P}$ in the vicinity of P . We render the image $\hat{I}$, depth $\hat{D}$, and features $\hat{\mathbf{F}}^{3D}$ from pose $\hat{P}$ and extract the 2D feature map $\hat{\mathbf{F}}^{2D}$ from the rendered image. Then, for a randomly sampled set of pixels u in I , we backproject it using D and re-project it into $\hat{I}$, yielding the pixel v in the rendered image. Then, we backproject v into 3D using $\hat{D}$ and re-project it into I . If the pixel distance between u and the reprojected v fall within a small threshold, (u, v) is considered as a valid correspondence, otherwise it is rejected.

3.3. Feature-based Localization, GSFFs-PR Feature

After training, the feature space along with the 3D Gaussians can be used for pose refinement by finding the pose that minimizes feature-metric errors between 2D extracted query features and 3D features rendered from the aforementioned pose. In particular, given a query image whose pose is unknown, we first extract its 2D feature map $\mathbf{F}^{2D}$ with our trained encoder. Second, we retrieve the closest database image with a global descriptor (our method is agnostic to the global descriptor used) and use the pose P_{init} of the retrieved image as initialization. From P_{init} , we render the $\mathbf{F}^{3D}$ feature map from GSFFs. Feature inconsistencies between the two feature maps are iteratively minimized with regard to the pose:

$$P^* = \min_{P \in SE(3)} \|\mathbf{F}^{2D} - \mathbf{F}^{3D}(P, \mathcal{G})\|_2^2, \quad (4)$$

where the pose is parametrized on $SE(3)$ and updates are done on the Lie algebra $se(3)$ by backpropagating explicitly through the rasterizer [43] (that we modify to feature-metric

	Model	Chess	Fire	Heads	Office	Pumpkin	Redkitchen	Stairs
SBM	HLoc [56]	0.8/0.11/100	0.9/0.24/99	0.6/0.25/100	1.2/0.20/100	1.4/0.15/100	1.1/0.14/99	2.9/0.80/72
	DSAC*[4]	0.5/0.17/100	0.8/0.28/99	0.5/0.34/100	1.2/0.34/98	1.2/0.28/99	0.7/0.21/97	2.7/0.78/92
	ACE [6]	0.5/0.18	0.8/0.33	0.5/0.33	1.0/0.29	1/0.22	0.8/0.2	2.9/0.81
	ACE + GSLoc [39]	0.5/0.15	0.6/0.25	0.4/0.28	0.9/0.26	1.0/0.23	0.7/0.17	1.4/0.42
RBM	NeFeS [13] (DFNet [12])	2/0.79	2/0.78	2/1.36	2/0.60	2/0.63	2/0.62	<u>5/1.31</u>
	MCLoc [70]	2/0.8	3/1.4	3/1.3	4/1.3	5/1.6	6/1.6	6/2.0
	SSL-Nif [53] (DV)	1/0.22/93	0.8/0.28/91	0.8/0.49/71	1.7/0.41/81	<u>1.5/0.34/86</u>	2.1/0.41/75	6.5/0.63/49
	NeRFMatch [91] (DV)	0.9/0.3	1.3/0.4	1.6/1.0	3.3/0.7	3.2/0.6	1.3/0.3	7.2/1.3
	GSplatLoc [64] (GSplaLoc)	0.43/0.16	1.03/0.32	1.06/0.62	1.85/0.4	1.8/0.35	2.71/0.55	8.83/2.34
	GSLoc [39] (DFNet)	0.7/0.20	0.9/0.32	0.6/0.36	<u>1.2/0.32</u>	1.3/0.31	<u>0.9/0.25</u>	2.2/0.61
	GSFFs-PR Feature (DV)	0.4/0.19/95	0.6/0.26/98	0.5/0.36/94	1.0/0.31/97	1.3/0.38/85	0.6/0.23/93	25.1/0.63/32
	GSFFs-PR Privacy (DV)	0.8/0.28/96	0.8/0.33/94	1.0/0.67/90	1.5/0.51/90	2.0/0.50/78	1.2/0.33/85	28.2/0.98/29

Table 1. **Localization results on 7Scenes** using SfM-based pseudo GT poses from [5]. We compare our results with Structure-Based Methods (SBM) and Rendering-Based Methods (RBM). Median position error (cm.) (↓)/ median rotation error (°) (↓)/ recall at 5cm/5° (%) (↑), when available. **Bold** are the best results, underline the second best.

refinement). At each refinement iteration, the pose is updated and features are rendered from this updated pose. We minimize the Eq. (4) objective until convergence.

4. Privacy-Preserving GSFFs

Due to the clustering and subsequent prototypical formulation adopted in GSFFs, we can seamlessly extend our model to be privacy-preserving. We follow the privacy definition from the literature [52, 75, 90] where privacy is refers to the inability for an attackers to recover texture/color information and fine-level details. Similarly, following prior works, we consider that having coarse geometric information does not violate privacy. Therefore, inspired by [52], we replace the high-dimensional features $\mathbf{g}_i$ that contain a lot of privacy-critical information with a single segmentation label. After training, the feature field, prototypes, and color information are removed.

Optimizing the segmentations. Features may be converted to segmentations by soft assigning them to the set of prototypes. Any Gaussian feature $\mathbf{g}_i$ or encoder feature $\mathbf{f}_i$ can be assigned to a prototype $\mathbf{p}_k$ by the likelihood of the feature belonging to the cluster k defined by $l_{ik}^{3D} = \exp(\mathbf{g}_i^\top \mathbf{p}_k) / \sum_{k'} \exp(\mathbf{g}_i^\top \mathbf{p}_{k'})$. The resulting scores form a pseudo-logits vector $\mathbf{l}_i$ that can be rendered with alpha-blending (*c.f.* Eq. (1)), where the color is replaced by the scores. Similarly, encoder features can be soft-assigned via $l_{ik}^{2D} = \exp(\mathbf{f}_i^\top \mathbf{p}_k) / \sum_{k'} \exp(\mathbf{f}_i^\top \mathbf{p}_{k'})$.

While the prototypical learning framework induces relatively well aligned encoder/rendered soft assignments, the localization accuracy can further be improved by learning to directly predict the 2D segmentations. Therefore, during training, we add and train a shallow classification head on top of the encoded features that learns to output the segmentation map $\mathbf{S}^{2D}$ directly from the images. To learn the segmentation head and to further refine our features in a self-supervised way, we add the following cross-entropy loss to

Eq. (2) and Eq. (3):

$$L_{CE} = - \sum_{u \in I} \mathbb{1}_u \cdot (\log(\mathbf{S}_u^{2D}) + \log(\mathbf{S}_u^{3D})) \quad , \quad (5)$$

where $\mathbf{S}^{3D}$ denotes the segmentation map with rendered pseudo-logits l_{ik}^{3D} , $\mathbb{1}_u$ is the one hot vector corresponding to the prototype label associated to the pixel u , using the exact same associations as in the prototypical contrastive loss encouraging consistency between the features and segmentations.

Privacy-preserving localization, GSFFs-PR Privacy.

After training, any potentially privacy-sensitive information is removed from the 3D Gaussian model (spherical harmonic, features, and GSFFs) as neither photometric information nor features are required for the privacy-preserving localization. The pseudo-logits obtained from the soft-assignments still contain too much information [52]. Therefore, we instead store hard assignments in the form of a single label per Gaussian $\mathcal{G}_i$, $k^* = \arg\max_k (\mathbf{l}_{ik})$. After this labeling procedure, the triplane GSFFs feature field and the prototypes are subsequently removed. As such the 3D models contain only geometric (Gaussians without color information) and segmentation information (cluster labels), effectively increasing the level of privacy. Furthermore, storing only the cluster labels makes the storage of the 3D representation orders of magnitudes smaller compared to storing features. Given a pose P , segmentation maps can be rendered by alpha-blending (via Eq. (1)) from one-hot vectors $\mathbb{1}_k$ (corresponding to label k), yielding $\mathbf{S}_P^{3D}$. Given an image, 2D segmentation maps $\mathbf{S}^{2D}$ are directly obtained through the segmentation head. The localization pipeline is similar to the one in Sec. 3.3 with the difference that we minimize segmentation inconsistencies between the segmentation labels instead of features:

$$P^* = \min_{P \in SE(3)} CE(\mathbf{S}^{2D}, \mathbf{S}_P^{3D}) \quad . \quad (6)$$

Model	King's	Old	Shop	St. Mary's
SBM	HLoc [56]	11 / 0.20	15 / 0.31	4 / 0.2
	Kapture R2D2 [?]]	5 / 0.10	9 / 0.20	2 / 0.10
	DSAC*[4]	15 / 0.30	21 / 0.40	5 / 0.30
	ACE [6]	29 / 0.38	31 / 0.61	5 / 0.30
	ACE + GSLoc [39]	25 / 0.29	26 / 0.38	5 / 0.23
PPM	GoMatch [90]	25 / 0.64	283 / 8.14	48 / 4.77
	DGC-GNN [75]	18 / 0.47	75 / 2.83	15 / 1.57
	SegLoc [52] (DV)	24 / 0.26	36 / 0.52	11 / 0.34
	GSFFs-PR Privacy (DV)	24 / 0.39	26 / 0.49	5 / 0.27
	CROSSFIRE [46] (DV)	47 / 0.7	43 / 0.7	20 / 1.2
RBM	NeFeS [13] (DFNet)	37 / 0.62	55 / 0.9	14 / 0.47
	MCLoc [70]	31 / 0.42	39 / 0.73	12 / 0.45
	SSL-Nif [53] (DV)	32 / 0.40	30 / 0.49	8 / 0.36
	NeRFMatch [91] (DV)	13 / 0.2	21 / 0.4	8.7 / 0.4
	GSLoc [39] (DFNet)	26 / 0.34	48 / 0.72	10 / 0.36
	GSplaLoc [64]	27 / 0.46	20 / 0.71	5 / 0.36
	GSFFs-PR Feature (DV)	18 / 0.27	21 / <u>0.41</u>	<u>4 / 0.26</u>
	GSFFs-PR Feature (tuned) (DV)	<u>17 / 0.26</u>	18 / 0.36	4 / 0.25
				8 / 0.26

Table 2. Localization results on Cambridge Landmarks. Comparison with Rendering-Based Methods (RBM), Privacy-Preserving Methods (PPM) and Structure-Based Methods (SBM). Median position error (cm.) ($\downarrow$), median rotation error ($^\circ$) ($\downarrow$). **Bold**: best results per method category, underline: second best. (tuned) stands for better hyperparameter tuning for this outdoor dataset.

5. Experimental Evaluation

In this section, we first provide details regarding the training and evaluation of GSFFs-PR. Then, in Sec. 5.1 we evaluate our feature-based VL pipeline (dubbed **GSFFs-PR** Feature) and our privacy-preserving VL pipeline (dubbed **GSFFs-PR** Privacy) on multiple real world datasets. We show that GSFFs-PR in general is more accurate than the respective non-privacy and privacy-preserving baselines. Sec. 5.2 ablates and discusses important aspects of our approach.

Experimental details. We train and optimize one model per scene for 50K iterations on a single NVIDIA A100 GPU. During the first 15K iterations only the Gaussians parameters are updated through the photometric loss. From iteration 15k, the triplane fields and the encoder are also optimized with the joint loss $L = L_{PHO} + .5L_{NCE} + .5L_{PRO} + .5L_{CE} + .1L_{TVL} + 0.05L_{Depth}$, where L_{TVL} is a total variation loss applied to the triplane parameters for regularization and L_{Depth} is a depth prior when available. To increase the convergence basin of our method, we define a coarse feature/segmentation space and a fine feature/segmentation space (both are trained in a similar fashion). The coarse triplane has a resolution of $R = 256$ and the fine triplane a resolution of $R = 1024$. Coarse image-based features are processed through a ViT [49] with patch size of 14, while fine image-based features are not down-sampled. The feature dimension d is set to 16 for both fine and coarse levels and the number of cluster K to 34 both for fine and coarse levels. A query image estimated pose is first iteratively refined with the coarse-level features or segmentations, then it is refined with the fine-level features or segmentations.

We evaluate our localization pipeline on the 7Scenes (with Depth SLAM pseudo ground truth poses [63] and with

Model	Chess	Fire	Heads	Office	Pumpkin	Redkitchen	Stairs
GoMatch [90]	4/1.65	13/3.86	9/5.17	11/2.48	16/3.32	13/2.84	89/21.12
DGC-GNN [75]	3/1.43	5/1.77	4/2.95	6/1.61	8/1.93	8/2.09	71/19.5
SegLoc [52]	4.5/1.02	4.0/1.51	4.1/1.80	4.7/1.33	6.3/1.88	6.0/1.87	43.7/8.27
GSFFs-PR	2.9/0.92	2.3/0.91	1.5/1.11	4.0/1.28	5.7/1.48	5.1/1.66	28.3/2.39

Table 3. Localization results on 7Scenes (DSLAM based pseudo GT), with median position error (cm.) ($\downarrow$)/ median rotation error ($^\circ$) ($\downarrow$). We compare **GSFFs-PR** (Privacy) with other privacy-preserving methods. Both SegLoc and **GSFFs-PR** refine the pose from DenseVLAD [69] initialization. Best numbers in **bold**.

SfM-based pseudo ground truth poses [5]), Indoor6 [18], and Cambridge Landmarks [32] real-world datasets. Results on the 12Scenes [71] dataset and comparisons against [84, 85] are provided in the Supplementary. The median position error (cm) and median rotation error ($^\circ$) are reported. Additionally, we report the recall at 5cm/5 $^\circ$ for the indoor scenes. Training parameters (loss coefficients, temperature, 3DGS hyperparameters) are kept the same for all datasets. More details are provided in the Supplementary.

5.1. Visual Localization

Comparison with feature rendering-based approaches.

We primarily compare GSFFs-PR Feature against other non-privacy-preserving rendering-based VL methods, which either use NeRFs [13, 53, 91] or 3DGS [39, 64] as their underlying 3D representations. These methods all require an initial pose estimate, which is obtained via pose retrieval with a global descriptors. More accurate global descriptors lead to more accurate initial pose estimates, which in turn makes refinement easier. For the sake of fairness, we try to use similar or less accurate global descriptors as the other baselines. The global descriptor used is specified between brackets. We report 7Scenes results (with SfM pseudo GT) in Tab. 1, and Cambridge Landmarks results in Tab. 2.

From Tab. 1, we observe that our **GSFFs-PR** Feature (DV) is best on 4 out of 7 scenes and second for two further ones in spite of relying on a less accurate initialization, *i.e.*, DenseVLAD is less accurate than DFNet or ACE. On Stairs, the Opacity Gaussian Field struggles due to its textureless and flat layout, yielding a poor results for the initial 3D structure that directly affects GSFFs-PR's refinement framework. From Tab. 2, we can see that GSFFs-PR is more accurate than other 3D Gaussian rendering-based baselines. Overall, our feature pipeline even displays competitive performances on some scenes compared to state-of-the-art structure-based methods.

Comparison with privacy-preserving approaches.

We compare the performance of the GSFFs-PRPrivacy version against other privacy-preserving visual localization methods, including SegLoc [52] and descriptor-less visual localization methods [75, 90], on Cambridge Landmarks in Tab. 2 and on 7Scenes with DSLAM pseudo GT in Tab. 3. Similar to our approach, both baseline try to remove the

	Model	scene1	scene2a	scene3	scene4a	scene5	scene6
NPPM	DSAC* [4]	12.3/2.06/18.7	7.9/0.9/28.0	13.1/2.34/19.7	3.7/0.95/60.8	40.7/6.72/10.6	6.0/1.40/44.3
	NBE+SLD [18]	6.5/0.9/38.4	7.2/0.68/32.7	4.4/0.91/53.0	3.8/0.94/ 66.5	6.0/0.91/40.0	5.0/0.99/50.5
	GSFFs-PR Feature (34 classes) (Sgl)	4.9/0.68/51	3.1/0.25/68	3.0/0.58/69	5.1/0.85/49	6.3/0.84/41	3.3/0.69/62
	GSFFs-PR Feature (84 classes) (Sgl)	3.6/0.5/63	2.7/0.2/77	2.4/0.39/73	3.6/0.54/63	4.8/0.53/53	2.5/0.42/68
PPM	SegLoc (Sgl) [52]	3.9/0.72/51.0	3.2/0.37/56.4	4.2/0.86/41.8	6.6/1.27/33.84	5.1/0.81/43.1	3.5/0.78/34.5
	GSFFs-PR Privacy (34 classes) (Sgl)	10.2/1.8/28	5.8/0.49/42	5.8/1.2/45	6.5/1.2/40	13.9/2.32/23	6.5/1.5/46
	GSFFs-PR Privacy (84 Classes) (Sgl)	4.6/0.67/53	4.1/0.33/49	4.1/0.74/55	4.0/0.61/58	6.6/0.92/38	3.7/0.75/57

Table 4. Localization results on Indoor6, comparison with Non-Privacy-Preserving Methods (NPPM) and Privacy-Preserving Methods (PPM). Median position error (cm.) ($\downarrow$)/ median rotation error ($^\circ$) ($\downarrow$)/ recall at 5cm/5 $^\circ$ (%) ($\uparrow$).

	KC (F)	OH (F)	KC (P)	OH (P)
Coarse only	37.5/0.80	670/1.10	92.2/1.55	82.5/1.50
Fine only	22.6/0.32	55.1/0.77	28.0/0.45	151/0.96
k-means	25.2/0.33	27.6/0.50	37.2/0.64	52.3/0.84
No SOpt	23.5/0.30	23.4/0.45	31.0/0.45	30.8/0.57
No MVR	20.0/0.28	22.5/0.49	29.6/0.51	29.1/0.49
No SE	23.5/0.31	21.9/0.42	27.0/0.42	26.0/0.47
GSFFs-PR	17.9/0.27	21.4/0.41	24.3/0.39	25.6/0.49

Table 5. Ablation of different component of our pipeline on King’s College (KC) and Old Hospital (OH) for Feature (F) and Privacy (P) versions. Median position error (cm.) / rotation error ($^\circ$) ($\downarrow$).

reliance on traditional high-dimensional features for visual localization and hence try to improve privacy. On both datasets, our pipeline proves to be much more accurate than all the other privacy-preserving methods. Finally, Tab. 4 shows results on the challenging Indoor6 dataset, which contains scenes with multiple rooms, uneven camera distributions, and large illumination changes. GSFFs-PR trained with 84 classes outperforms SegLoc in average recall, while GSFFs-PR trained with 34 classes is on par with SegLoc. This shows that increasing the number of classes increases the discriminative power of the segmentation, which is particularly beneficial for larger and complex scenes such as *scene1* and *scene5*. The Supplementary provides an ablation study on varying the number of classes.

5.2. Ablations and Discussion

Tab. 5 shows results for our ablation studies, where we study a single component per ablation experiment.

No hierarchy. First, we evaluate the importance of the hierarchical approach by either removing the fine level (Coarse Only) or by removing the coarse level (Fine Only). The results in Tab. 5 (first two rows) show that alignment on the finer level yields better pose accuracy and that increasing the convergence basin with a hierarchical approach is necessary to ensure accurate localization.

Clustering. As shown in Tab. 5 (k-means), when we replace the spectral clustering with a simple K-means clustering to derive the prototypes in the 3D feature space, the accuracy drops, confirming that spectral clustering is a critical component of the representation learning process.

Feature field learning. The accuracy drops also when 1)

we replace the scale-aware feature encoding with a simple point-wise projection of 3D Gaussian centers (*c.f.* Tab. 5 (No SE)) or 2) when we remove the multi-view regularization in the contrastive losses (*c.f.* Tab. 5 (No MVR)). This shows that both components leverage the geometric information contained in the 3D Gaussian model, yielding more robust and more discriminative representations, and hence a higher pose accuracy.

Segmentation optimization. By removing the loss from Eq. (5), we do not optimize the segmentations during training (Tab. 5 (No SOpt)). Instead, during inference, we align soft assignments between features and prototypes. Due to the prototypical loss, the soft assignments are aligned well enough to provide decent accuracy, however the segmentation head provides a significant gain on accuracy. Furthermore, the feature alignment also benefit from the segmentation optimization during training.

Sparsity of the representation. To evaluate the role of the dense representation, we adopt an approach closer to SegLoc by only applying pose refinement on a sparse set of points. To do this we reproject only Gaussian centers into the image space, interpolate 2D extracted/3D rendered segmentations at these locations and apply Eq. (6) for pose refinement. In this sparse setup, the pose refinement does not converge, which suggests that a sparse signal is not sufficient when backpropagating through the renderer.

6. Conclusion

In this paper, we propose a Gaussian Feature Field (GSFFs) that combines an explicit geometry representation (3DGS) and an implicit feature field into a scene representation suitable for visual localization through pose refinement. The resulting VL framework relies on a learned feature embedding space shared between the scale-aware GSFFs and a 2D encoder. Both the field and the encoder are optimized in a self-supervised way via contrastive learning. By leveraging the geometry and structure of the GS representation, we regularize the training through multi-view consistency and clustering. The clusters are further used to convert the features into segmentations to enable privacy-preserving localization. Both the privacy-preserving and non-privacy-preserving variants of our localization framework outperform the relevant state-of-the-art.

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